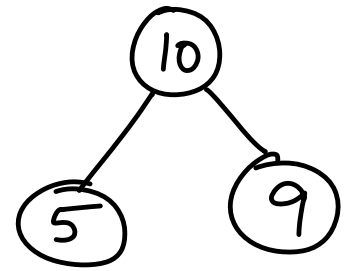
Heaps:

Max-Heap property :

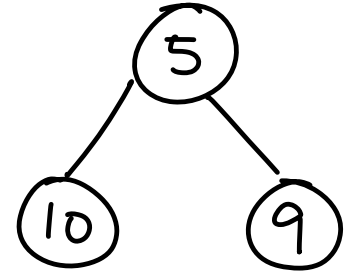
$$A[\text{Parent}(i)] \geq A[i]$$

eg

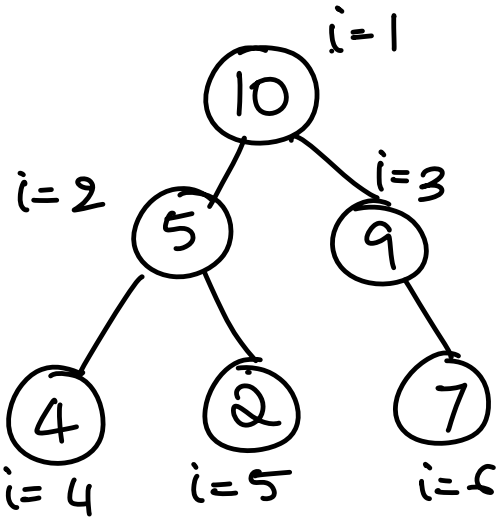


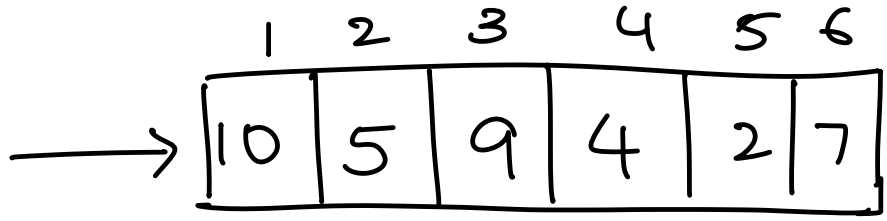
Min-Heap property :

$$A[\text{Parent}(i)] \leq A[i]$$



Heap $\longleftrightarrow$ Array :





Max-Heapify (A, i)

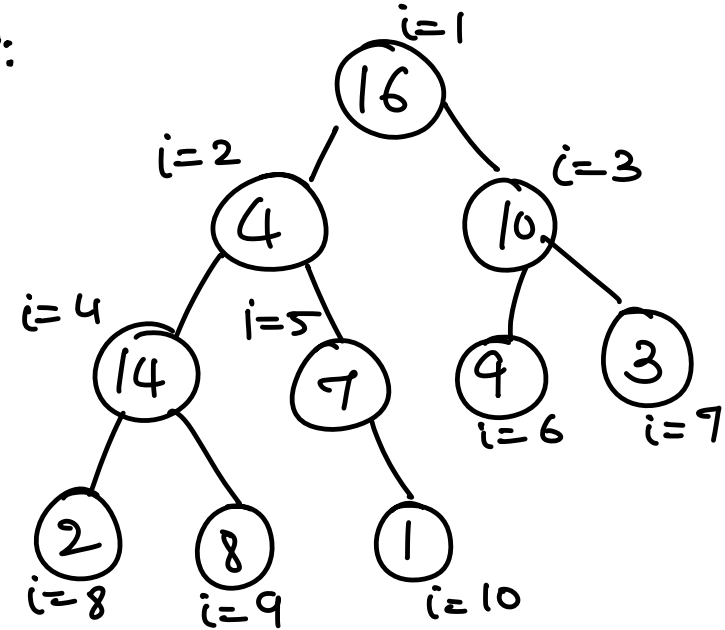
1. $l \leftarrow$ left(i) (or 2i)
2. $r \leftarrow$ right(i) (or 2i+1)
3. If $l \leq$ heapsize[A] and A[l] > A[i]
4. then largest $\leftarrow l$
5. else largest $\leftarrow i$
6. If $r \leq$ heapsize[A] and A[r] > A[largest]
7. then largest $\leftarrow r$

8. If largest ≠ i

9. then A[i] ⟷ A[largest]

10 Max-Heapify (A, largest)

eg:



A

i	1	2	3	4	5	6	7	8	9	10
A	16	4	10	14	7	9	3	2	8	1

Max-Heapify (A, 2)

heap-size (A) = 10.

$l = 4, \quad r = 5$

$l = 4 \leq 10$ and $A[\underset{l}{4}] = 14 > A[\underset{i}{2}] = 4$

$\Rightarrow$ largest $\leftarrow 4$

$r = 5 \leq 10$ and $A[\underset{r}{5}] = 7 \not> A[\underset{\text{largest}}{4}] = 14$

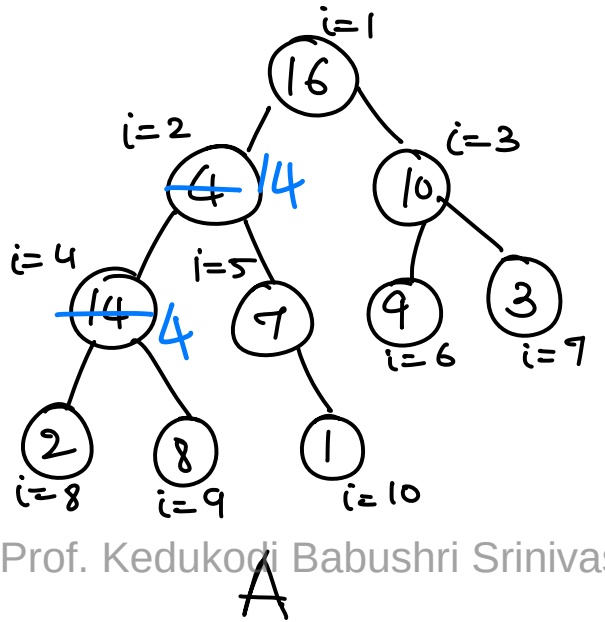
largest remains same

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largest $= 4 \neq i = 2$

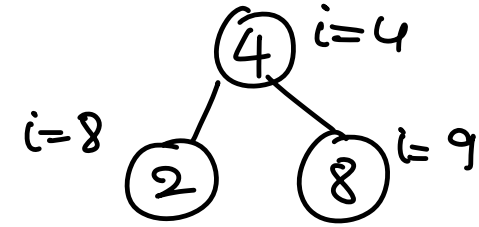
$A[\overset{i}{2}] \longleftrightarrow A[\overset{largest}{4}]$

$4 \longleftrightarrow 14$



Max-Heapify (A, largest 4)

Focus on



$l = 8, \quad r = 9.$

$l = 8 \leq 10$ and $A[\overset{l}{8}] \quad A[\overset{i}{4}]$

$2 \not> 4$

largest $\leftarrow 4 = i$

$r = 9 \leq 10$ and $A[\overset{r}{9}] \quad A[\text{largest}\ 4]$

$8 > 4$

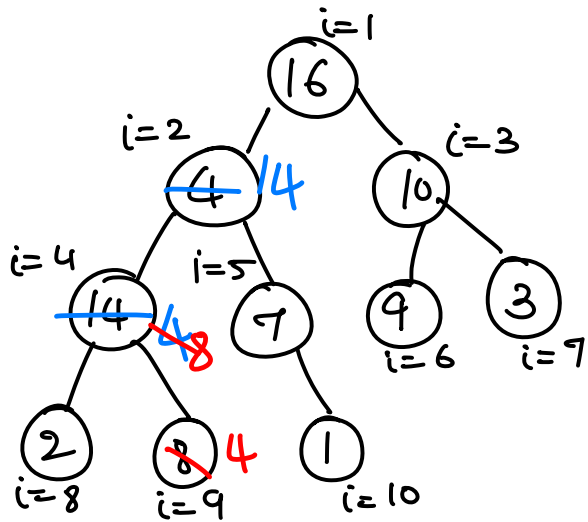
largest $\leftarrow r = 8$

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largest $\leftarrow 8 \neq i = 4$

$A[\underset{i}{4}] \longleftrightarrow A[\underset{largest}{9}]$

$4 \longleftrightarrow 8$



A

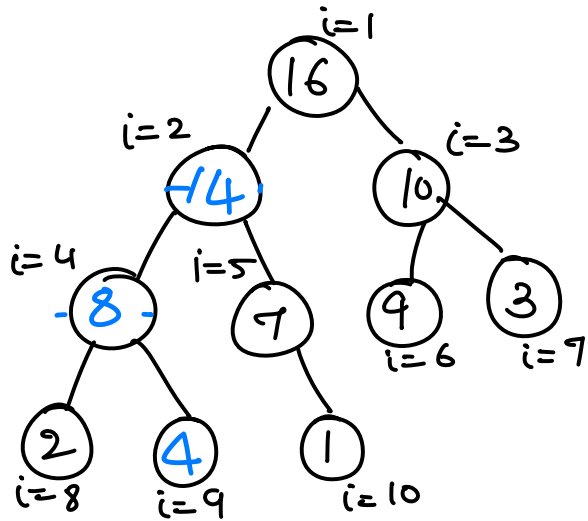
Max-Heapify (A, largest q)

$l = 18 \qquad r = 19$

$l = 18 \nleq 10$

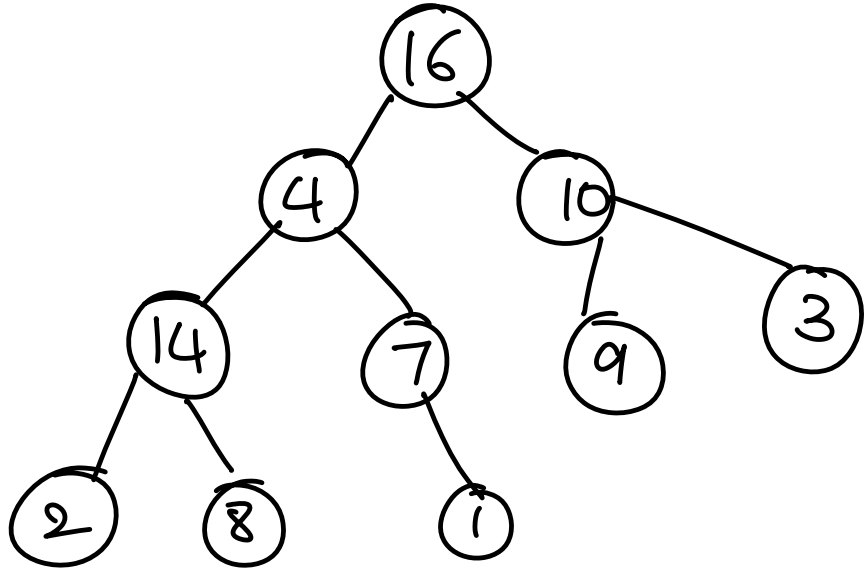
$r = 19 \nleq 10$. (Stops)

Final:



A

eg:



$i=1, \quad l=2, \quad r=3$

A[2] A[1]

$4 \not> 16$

A[3] A[10]

$10 \not> 16$

Min-Heapify (A, i)

1. $l \leftarrow$ left (i) (or 2i)
2. $r \leftarrow$ right (i) (or 2i+1)
3. If $l \leq$ heapsize [A] and A[l] < A[i]
4. then smallest $\leftarrow l$
5. else smallest $\leftarrow i$
6. If $r \leq$ heapsize [A] and A[r] < A[smallest]
7. then smallest $\leftarrow r$

8. If Smallest $\neq$ i

9. then A[i] $\longleftrightarrow$ A[Smallest]

10 Max-Heapify (A, Smallest)

Build-Max-Heap(A)

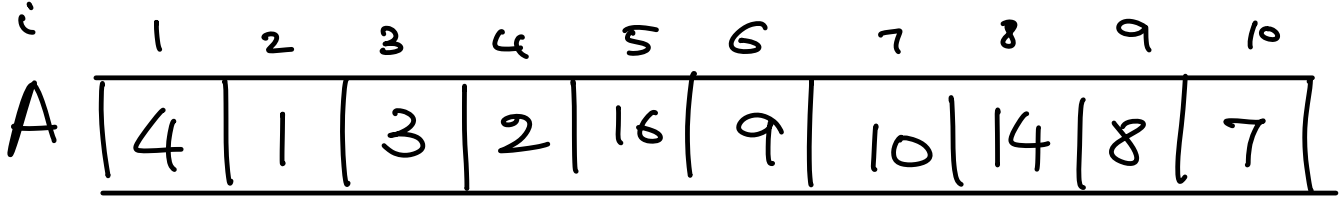
1. heap-size[A] $\leftarrow$ length[A]
2. For $i \leftarrow \left\lfloor \frac{\text{length}[A]}{2} \right\rfloor$ to 1
3. do Max-Heapify(A, i).

Heap - Extract-Max(A):

1. max ← A[1]
2. A[1] ← A[heap-size[A]]
3. heapsize[A] ← heapsize[A] − 1
4. Max-Heapify(A, 1)
5. return max.

Heap Sort (A):

1. Build-Max-Heap (A)
2. For $i \leftarrow$ length[A] to 2
3. $\quad A[1] \leftrightarrow A[i]$
4. $\quad$ heapsize[A] $\leftrightarrow$ heapsize[A] - 1
5. $\quad$ Max-Heapify (A, 1).



Heap Sort (A):

1. Build-Max-Heap(A).

heapsize[A] ← 10

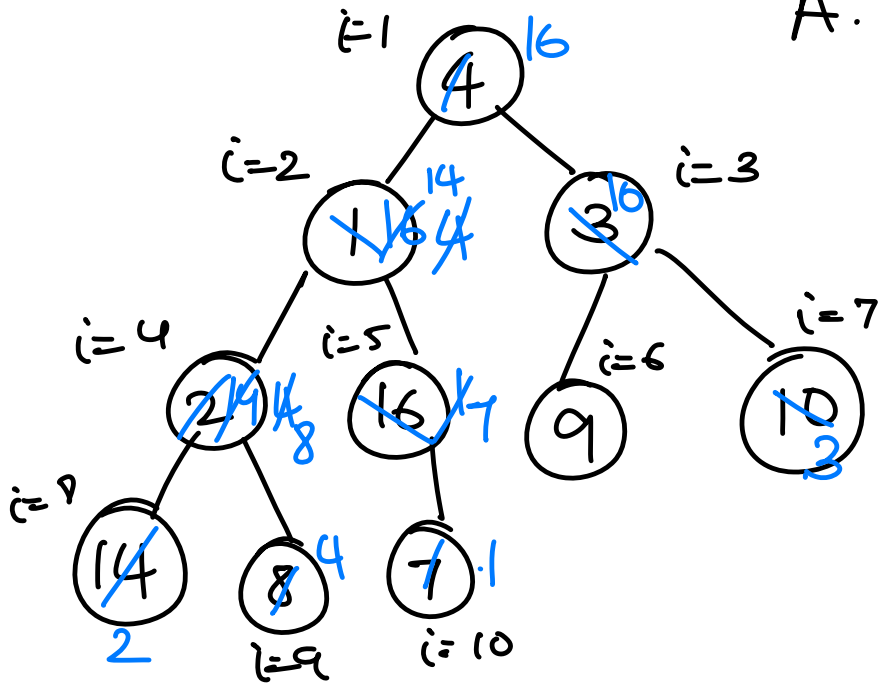
For $i \leftarrow \left\lfloor \frac{length(A)}{2} \right\rfloor$ to 1

5^{11} to 1

do Max-Heapify (A, i)

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A.



$i=5 \rightarrow$ remains same

$i=4 \rightarrow$ $14 \leftrightarrow 2$

$i=3 \rightarrow$ $10 \leftrightarrow 9$

$i = 2 \rightarrow$ $1 \leftrightarrow 16$

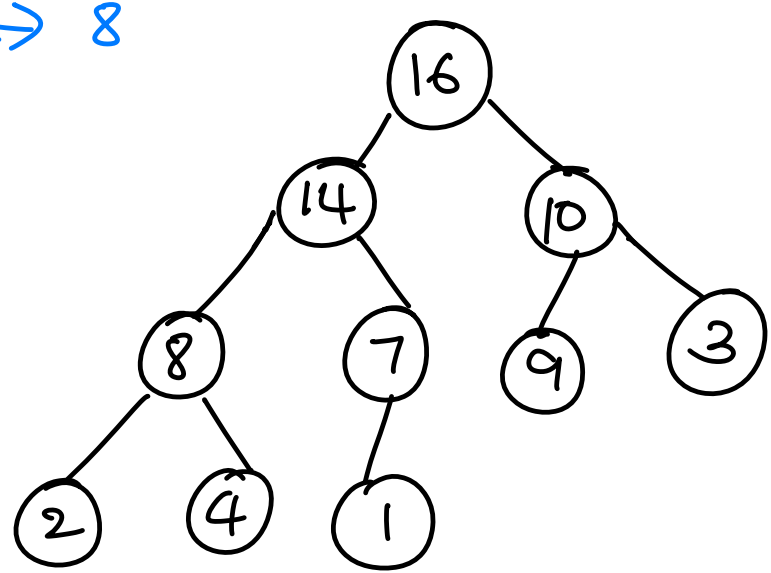
$1 \leftrightarrow 9$

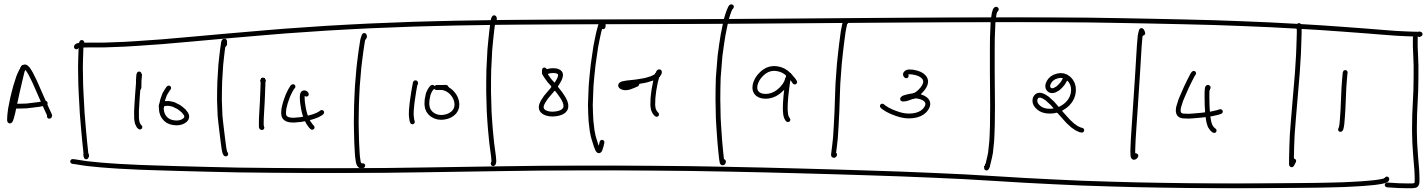
$i = 1 \rightarrow$ $4 \leftrightarrow 16$

$4 \leftrightarrow 14$

$4 \leftrightarrow 8$

Final :





For $i \leftarrow$ length[A] to 2

" 10 to 2

A[1] $\leftrightarrow$ A[i]

heapsize[A] $\leftarrow$ heapsize[A] $-$ 1

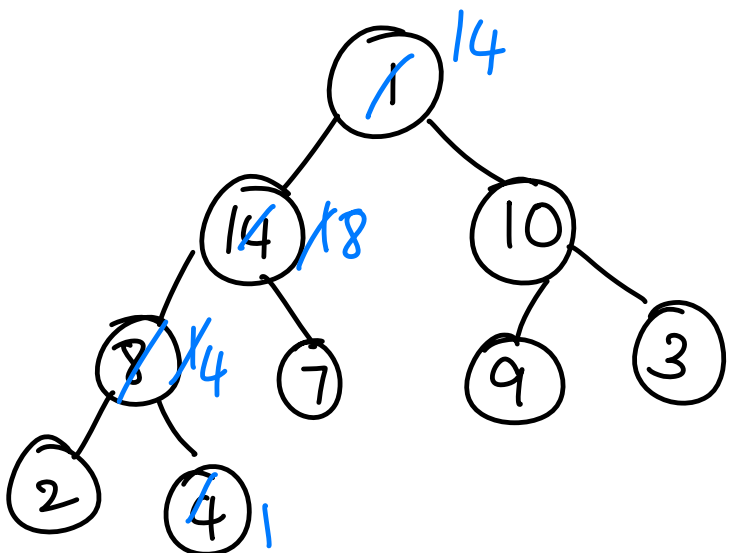
Max Heapify (A, 1).

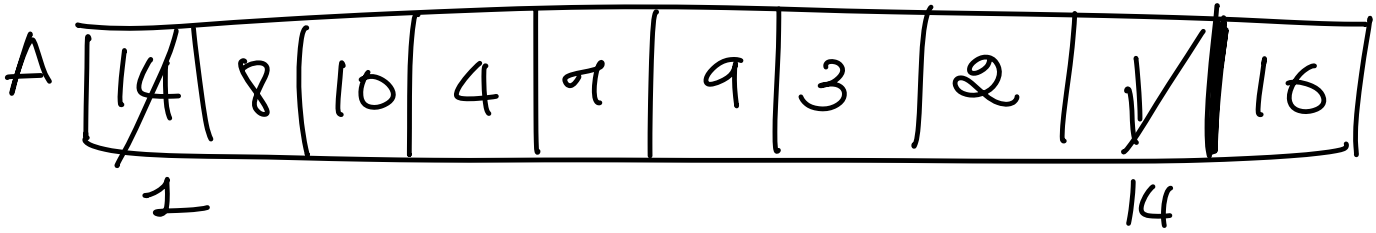
i = 10.

	1									
A	~~16~~	14	10	8	7	9	3	2	4	~~16~~

$A[1] \leftrightarrow A[0]$.

heapsize[A] $\leftarrow$ 9. Max-Heapify(A, 1).

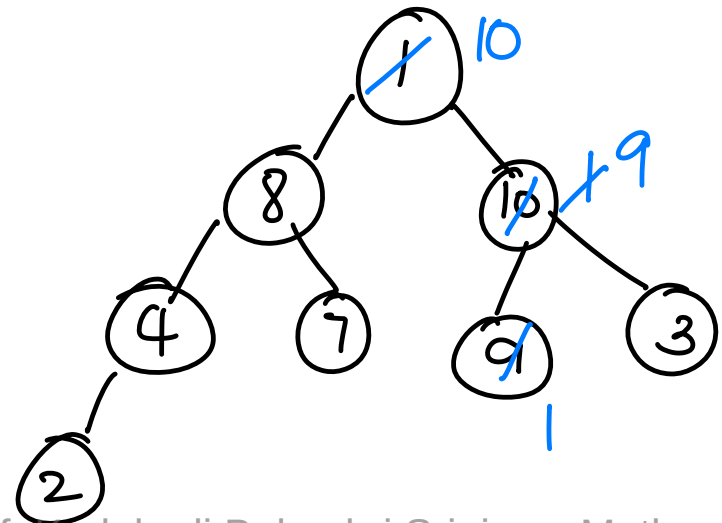


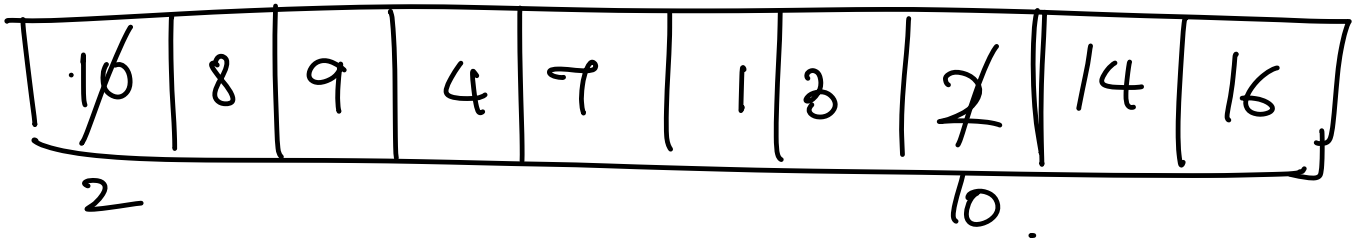


$i = 9$

$A[9] \leftrightarrow A[1]$ $\quad$ heapsize$[A] \leftarrow 8$

$16 \leftrightarrow 1$





$i = 8$

$A[8] \longleftrightarrow A[1]$